Ascend Coaching



EVERYTHING YOU NEED TO KNOW ABOUT STARTING PEPTIDES

What peptides are, what they actually do in your body, what the science says, and the questions to ask before you ever start one.

FREE STARTER GUIDE · 2026 EDITION

Educational information only, not medical advice. Peptides that are drugs should only be used under the care of a licensed medical provider. Information current as of November 2026.

CHAPTER 1

01 My experience with peptides

Before the science, here is what peptides have looked like in my own training.

HEAVIEST BULK	WHERE I FINISHED	TIME IT TOOK
180 lbs	**147 lbs**	**3 months**
At 5'7", the heaviest I have ever been.	At around 11% body fat.	With a proper diet and structured training the whole way.

Coming back from injuries

Like most people who have trained hard for years, I have dealt with my share of injuries. For my injuries I used what is often called the "Wolverine blend", a combination of BPC-157 and TB-500. In my own experience, my recovery felt up to 40% faster than I would normally expect, and I was able to get back to training sooner.

The bulk

I then went through a heavy bulk, pushing my bodyweight up to 180 lbs at 5'7", the heaviest I have ever been. The goal was to build as much muscle as possible, knowing I would have a lot of fat to lose afterward.

The cut: 180 to 147 in 3 months

To cut down, I used GLP-1s and tesamorelin alongside a proper diet and training. Over about three months I went from 180 lbs to a very aesthetic 147 lbs at around 11% body fat. The GLP-1s helped me control appetite while eating in a deficit, and I used tesamorelin to help sculpt and target stubborn fat.

> **The part people skip:** none of this worked on its own. Every week of that cut was built on hitting my calories and protein, training hard and consistently, and recovering properly. Peptides were a tool on top of the work, never a replacement for it.

This chapter is my personal experience, not a typical result and not a recommendation. Your results, risks and what is appropriate for you can be very different. Several of these compounds are not FDA-approved, and the approved ones require a prescription. Talk to a licensed medical provider before using any peptide. The rest of this guide explains what the science and the law actually say.

CHAPTER 2

02 What are peptides?

Peptides are short chains of amino acids, the same building blocks that make up protein. Think of them as small proteins.



A peptide is a string of amino acids joined by peptide bonds. Shorter chains are called peptides; long, folded chains are called proteins.

Peptides vs. proteins

Both are made of amino acids linked together. The difference is mostly size: peptides are short chains, while proteins are long chains that fold into complex 3D shapes. Because they are smaller, many peptides can act quickly as chemical messengers.

Your body already makes thousands of them

Scientists have identified roughly **7,000 naturally occurring peptides** in the human body. Many are hormones and signaling molecules you have heard of:

BLOOD SUGAR

Insulin

Moves sugar from your blood into your cells after you eat.

BONDING

Oxytocin

Involved in social bonding, childbirth and breastfeeding.

PAIN & MOOD

Endorphins

Your body's natural pain relievers, part of the "runner's high".

WATER BALANCE

Vasopressin

Helps your kidneys control how much water you keep.

APPETITE

GLP-1

Released from the gut after meals; slows digestion and signals fullness.

STRUCTURE

Collagen fragments

Pieces of the protein that makes up skin, tendons and connective tissue.

Key idea: "Peptide" describes a type of molecule, not a single product. Insulin, a collagen powder and an unregulated vial bought online can all be called peptides, but they are completely different in how well they are studied and regulated.

CHAPTER 3

03 What do peptides do?

Most peptides work as messengers. They carry a signal from one part of the body to another and tell cells what to do.

How the signal works

1. Released
A gland or tissue releases a peptide into the blood or nearby tissue (for example, the pancreas releases insulin after a meal).

2. Travels
The peptide moves to its target cells.

3. Binds a receptor
It locks onto a specific receptor on the cell surface, like a key in a lock.

4. Triggers a response
The cell changes what it is doing: absorbing sugar, releasing a hormone, repairing tissue, or sending its own signals.

5. Broken down
Enzymes break most natural peptides down quickly, often within minutes, so the signal is short-lived.

Why that matters for peptide products

- **Very specific targets.** Because each peptide fits particular receptors, peptide drugs can have precise effects. That is why they are popular in medicine.
- **Hard to take by mouth.** Your digestive system breaks peptides down like any other protein, so most peptide drugs are given by injection. (Oral semaglutide is a notable exception, made possible by a special absorption enhancer.)
- **Short-lived.** Natural peptides break down fast, so drug makers often modify them to last longer. Semaglutide, for example, lasts long enough for once-weekly dosing.
- **Effects ripple outward.** A peptide that raises one hormone can affect several others downstream, which is one reason side effects are not always obvious.

> **Bottom line:** Peptides are powerful precisely because they act as signals. That is also why taking one is not like taking a vitamin. You are changing your body's messaging system.

CHAPTER 4

04 What peptides are studied for

Peptides are researched for almost every goal people care about. The most common: skin and complexion (GHK-Cu), antioxidant support (glutathione), and building muscle (IGF-1 and growth hormone peptides). Here is what each popular peptide does, grouped by goal.

◆ Skin, hair and appearance

GHK-Cu (copper peptide)

Some evidence

A copper-binding peptide found naturally in blood that drops with age. One of the most studied peptides for skin: lab and topical studies show it supports collagen production, skin firmness and wound healing. Common in serums and creams.

Collagen peptides

Modest

Broken-down collagen taken as a powder or drink. Several small trials found modest improvements in skin elasticity and hydration, and some in joint comfort.

Melanotan II

Unapproved

Mimics the hormone that darkens skin, so it is used for tanning. Not approved anywhere; linked to nausea, flushing and changes in moles. A related drug, afamelanotide, is FDA-approved only for a rare light-sensitivity disorder.

◆ Antioxidant, immunity and sexual health

Glutathione

Mixed

A tiny three-amino-acid peptide and the body's main antioxidant. It helps neutralize free radicals and supports the liver's detox work. Popular for wellness and skin brightening; evidence for supplements and injections is mixed, and absorption from pills is debated.

Thymosin alpha-1

Limited

Supports immune function. Approved as a drug in some countries for certain infections and as an add-on for some cancers, but not in the U.S.

PT-141 (bremelanotide)

FDA-approved

Works on the brain's melanocortin receptors to increase sexual desire. FDA-approved in 2019 as Vyleesi for low sexual desire in premenopausal women; used off-label by men.

🔷 Building muscle: IGF-1 and growth hormone peptides

IGF-1

Rx for rare use

Insulin-like growth factor 1 carries out many of growth hormone's effects, including muscle and tissue growth. A prescription form is FDA-approved only for a rare growth disorder in children. Versions like IGF-1 LR3 are unapproved and banned in sport.

GH secretagogues

Limited

Ipamorelin, CJC-1295, sermorelin and GHRP-6 signal your pituitary gland to release more of your own growth hormone. Used for muscle, recovery, fat loss and sleep. Human evidence for muscle gain in healthy adults is limited, and all are banned in sport.

Tesamorelin

FDA-approved

A growth hormone releasing hormone analog. FDA-approved to reduce excess belly fat in people with HIV; used off-label for fat loss and body composition.

🔷 Fat loss and metabolism

Semaglutide, tirzepatide

FDA-approved

Copy the gut hormone GLP-1 (and GIP for tirzepatide) to reduce appetite and slow digestion. In large trials they produced roughly 15 to 20% average weight loss. Prescription only.

MOTs-C

Very limited

A peptide made by mitochondria that may help the body use glucose and fat. Promising in animal studies, but no identified human safety data per the FDA.

AOD-9604

Weak

A fragment of growth hormone marketed for fat loss. It was tested as an obesity drug, but development stopped after trials did not show meaningful weight loss.

🔷 Recovery and healing

BPC-157

Very limited

Based on a protein found in stomach juice. Studied for healing tendons, muscles, ligaments and the gut, with promising results almost entirely in animals.

TB-500

Very limited

A fragment of thymosin beta-4, a protein involved in cell movement and tissue repair. Used for injury recovery and flexibility; human data is lacking.

KPV

Very limited

A small anti-inflammatory fragment of a natural hormone. Being studied in the lab for gut inflammation and skin conditions.

🔷 Brain, sleep and longevity

Semax and Selank

Limited

Developed in Russia and used there as nasal sprays for focus, memory and anxiety. Very little research outside Russia.

DSIP

Very limited

Delta sleep-inducing peptide, named for its link to deep sleep in early studies. Results in people have been inconsistent.

Epitalon

Very limited

Marketed for anti-aging and longevity, based mostly on small Russian studies about telomeres and sleep cycles.

"FDA-approved" applies only to the approved use and form. Evidence labels reflect human research. Being studied for something is not the same as being proven to work for it.

CHAPTER 5

05 The three kinds of peptide products

When people say they are "starting peptides", they could mean three very different things. Knowing which one you are looking at is the most important first step.

1. FDA-approved peptide drugs

The FDA has approved roughly **100 peptide drugs** for specific medical conditions. Each went through clinical trials proving it is safe and effective for that use, and is made to pharmaceutical standards.

Examples: insulin (diabetes), semaglutide (Ozempic, Wegovy) and tirzepatide (Mounjaro, Zepbound) for diabetes and weight management, and tesamorelin (Egrifta) for a specific type of abdominal fat in people with HIV. All require a prescription.

2. Unapproved "wellness" or "research" peptides

These are the peptides most often discussed in gyms and online: BPC-157, TB-500, ipamorelin, CJC-1295, MOTs-C and others. **None of these is FDA-approved for any use.** Human research is limited, correct doses are not established, and many are sold online labeled "for research use only", a label sellers use to avoid drug rules even though they are being bought for human use.

3. Food and supplement peptides

Collagen peptides and other protein fragments from food are sold as dietary supplements. They are taken by mouth and are generally low risk, but supplements are not reviewed by the FDA for effectiveness before they are sold, and quality varies by brand.

APPROVED DRUGS

Strongest evidence

Clinical trials, pharmaceutical quality, prescription and medical supervision.

WELLNESS PEPTIDES

Weakest evidence

Mostly animal or lab studies, unknown doses, quality depends on the source.

SUPPLEMENTS

Mixed evidence

Low risk, modest effects; not FDA-reviewed for effectiveness.

CHAPTER 6

06 Popular peptides at a glance

What people use them for, how strong the human evidence is, and where each one stands legally and in sport.

Peptide	What it's used or marketed for	Human evidence	Status (Nov 2026)
Semaglutide / tirzepatide	Type 2 diabetes, weight management	Strong Large clinical trials	FDA-approved, prescription only
Insulin	Diabetes	Strong	FDA-approved, prescription
Tesamorelin	Approved for HIV-associated abdominal fat; marketed online for fat loss	Strong for its approved use only	FDA-approved (Rx); banned in sport (WADA S2)
Collagen peptides	Skin, joints, connective tissue	Modest Mixed, small trials	Dietary supplement
GHK-Cu	Skin and hair (topical), "healing" (injected)	Some for topical skin use; Very limited injected	Cosmetic topically; injectable not FDA-approved
BPC-157	Tendon, muscle and gut healing	Very limited Mostly animal studies	Not FDA-approved; banned in sport (WADA S0)
TB-500 (thymosin beta-4 fragment)	Injury recovery, flexibility	Very limited No identified human safety data per FDA	Not FDA-approved; banned in sport (WADA S2)
Ipamorelin, CJC-1295, GHRP-6	Raising growth hormone for recovery, body composition	Limited	Not FDA-approved; banned in sport (WADA S2)
MOTs-C	Metabolism, endurance	Very limited No identified human safety data per FDA	Not FDA-approved

"Banned in sport" refers to the World Anti-Doping Agency 2026 Prohibited List, which applies to athletes in tested competition. S0 = non-approved substances; S2 = peptide hormones, growth factors and related substances. These are banned at all times, in and out of competition.

Reading this table: "Very limited" does not mean a peptide doesn't work. It means nobody has done the human studies to show whether it works, what dose is right, or what the long-term risks are.

CHAPTER 7

07 Where the law stands in 2026

Peptide rules have been changing fast. Here is what has actually happened, in plain English.

2023
The FDA restricted compounding pharmacies from making several popular peptides, including BPC-157, citing possible immune reactions, impurity concerns, and little or no safety information.

April 15, 2026
The FDA announced it would remove 12 peptides from its "Category 2" list of substances with significant safety concerns. The list included BPC-157, TB-500, MOTs-C, Semax, Epitalon, KPV, DSIP, LL-37, Melanotan II and injectable GHK-Cu.

July 2026
The FDA's Pharmacy Compounding Advisory Committee was scheduled to review several of these peptides. Its recommendations are not binding, and the FDA must still complete formal steps before anything changes.

What this does and doesn't mean

- **Removal from the "safety concerns" list is not FDA approval.** None of these peptides has been proven safe and effective in clinical trials.
- **Being off that list does not automatically make them legal to compound.** They remain outside the FDA's compounding enforcement policy unless and until they are formally added.
- **"Research use only" products are not meant for people.** They are not made to pharmaceutical standards, and the FDA has taken action against companies marketing them for human use.

> As one expert put it to Live Science: **"'Allowed to be compounded' does not mean 'FDA-approved.'"**

CHAPTER 8

08 The real risks

Most problems with peptides don't come from the idea of peptides. They come from unknown products, unknown doses, and no medical oversight.

PURITY

What's actually in the vial?

Gray-market products can be under-dosed, over-dosed, contaminated, or a different substance entirely.

IMMUNE REACTIONS

Immunogenicity

The FDA flagged that impurities and clumped peptides can trigger immune responses.

STERILITY

Infection risk

Injecting a non-sterile product can cause infections at the injection site or beyond.

DOSING

No established doses

For unapproved peptides, there is no tested dose for humans. Online "protocols" are guesswork.

INTERACTIONS

Stacking unknowns

Little is known about combining peptides with each other or with medications.

LONG TERM

Unknown effects

Peptides that raise growth hormone or growth factors have unknown long-term effects.

Approved drugs have risks too

Even well-studied peptides have side effects. GLP-1 medications like semaglutide commonly cause nausea and digestive issues, and can lead to muscle loss along with fat loss if protein intake and strength training are neglected. This is exactly why they are prescribed and monitored by a medical provider.

Athletes: one more risk

If you compete in any drug-tested sport, many peptides (including BPC-157, TB-500, and growth hormone secretagogues) are banned at all times under the WADA Prohibited List.

REFERENCES

Sources

1. Zieba, J. "Peptides are the latest wellness fad. What are they, and are they safe?" Live Science, August 18, 2026. livescience.com/health/medicine-drugs/peptides-are-the-latest-wellness-fad-what-are-they-and-are-they-safe
2. U.S. Food and Drug Administration. "Safety Risks Associated with Certain Bulk Drug Substances Nominated for Use in Compounding." Content current as of April 22, 2026. fda.gov
3. Frier Levitt. "FDA to Remove 12 Popular Peptides from the Category 2 'Do Not Compound' List." April 16, 2026. frierlevitt.com/articles/fda-peptides-do-not-compound-list-update-2026
4. World Anti-Doping Agency. "The 2026 Prohibited List: International Standard." Effective January 1, 2026. Sections S0 and S2.

Want help with the fundamentals?

At **Ascend Coaching**, a real personal trainer with 9 years of experience builds your training and nutrition plan 1:1, and adjusts it as you progress. Start your ascension today at Ascend Coaching.

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